SOCIAL MEDIA ANALYTICS (MYSQL)

1 Project Overview

This project focuses on designing and analyzing a **Social Media Database System** using SQL. The system simulates a real-world social media platform where users can create posts, follow others, like posts, comment, and use hashtags.The project also performs analytics to extract meaningful insights such as:

- Top influencers
- User engagement
- Trending hashtags
- Post activity patterns
- Follower growth

2 Project Objective

- To design a structured social media database using SQL
- To implement relationships between tables (users, posts, followers, likes, comments, hashtags)
- To apply SQL concept**s** like:

  Joins

  Group By & Having

  Subqueries

  Functions & Stored Procedures

  Views & Triggers
- To analyze user behavior and engagement through:

  Top active user

  Most liked/comented posts

  Trending hashtags

  Follower growth

3 Data source

The data used in this project is created to simulate a real-world social media platform. It is generated and inserted manually using SQL INSERT statements. The dataset consists of multiple interconnected tables representing different components of a social media system:

- **Users Table**

  Contains user details such as user ID, username, and email.

- **Posts Table**

  Stores user-generated content including captions and posting timestamps.
- **Followers Table**

  Represents relationships between users (who follows whom).
- **Likes Table**

  Captures which users liked which posts.
- **Comments Table**

  Stores user comments on posts along with timestamps.
- **Hashtags Table**

  Contains predefined hashtags used for categorizing posts.
- **Post_Hashtags Table**

  A mapping table that links posts with hashtags (many-to-many relationship).
- **Notifications Table**

  Stores system-generated notifications triggered when a post receives a like.

4 Problem Statement

- Design a database system that can efficiently store social media data
- Ensure data consistency and relationships between entities
- Analyze the data to answer key questions like:

  Who are the most influential users?

  Which posts are highly engaging?

  What hashtags are trending?

  How active are users over time?

  Which users are gaining more followers?

5 Attributes (Tables & Columns Details)

USERS TABLE

Attribute	Type	Description
User_id	INT	Unique user ID

User_name	VARCHAR	User name
email	VARCHAR	Unique email

POSTS TABLE

Attribute	**Type**	**Description**
Post_id	INT	Unique post id
User_id	INT	User who created post
caption	TEXT	Post content
Posted_at	DATETIME	Post time

FOLLOWERS TABLE

Attribute	**Type**	**Description**
Follower_id	INT	Follower user
Following_id	INT	Followed user
Follow_date	DATETIME	Follow timestamp

COMMENTS TABLE

Attribute	**Type**	**Description**
Comment_id	INT	Unique comment ID
Post_id	INT	Related post
User_id	INT	Comment author
Comment_text	TEXT	Comment content
Commented at	DATETIME	Comments time

LIKES TABLE

Attribute	**Type**	**Description**
Like_id	INT	Unique like ID
Post_id	INT	Liked post
User_id	INT	User who liked
Liked_at	DATE	Liked time

HASHTAGS TABLE

Attribute	Type	Description
Tag_id	INT	Unique hashtag ID
Tag_name	VARCHAR	Hashtag name

POST_HASHTAGS TABLE

Attribute	Type	Description
Post_id	INT	FK → posts
Tag_id	INT	FK → hashtag

6 Data Processing (SQL Operations)

□ Data Creation

- Tables created using CREATE TABLE
- Constraints:

  Primary Key

  Foreign Key

  Unique

  Default

✓ Data Insertion

- Data was inserted into all tables (users, posts, followers, likes, comments, hashtags, post_hashtags) using SQL INSERT statements
- Ensured proper mapping between tables using **foreign keys**

✓ Data Cleaning & Validation

- Checked for NULL values (e.g., captions in posts)
- Verified unique constraints (usernames, emails)
- Ensured valid relationships between tables (no invalid user_id or post_id)

✓ Data Filtering

- Used WHERE, LIKE, IN, and BETWEEN to filter:

  Users based on name patterns

Posts within specific date ranges

Emails by domain

✓ Data Transformation

- Applied SQL functions such as:
  - UPPER() → to standardize usernames
  - MONTHNAME() → to extract month from dates
  - LENGTH() → to analyze caption size

✓ Aggregation

- Used COUNT() to calculate:
  - Number of posts per user
  - Total likes and comments
- Applied GROUP BY and HAVING to summarize and filter grouped data

✓ Joins for Data Integration

- Combined multiple tables using:
  - INNER JOIN
  - LEFT JOIN
- Generated meaningful datasets like:

  - Post details with user info
  - Total likes and comments per post

✓ Subqueries

- Used subqueries to:
  - Compare follower counts with average
  - Identify high engagement posts

✓ Stored Procedures & Functions

- Created:
  - **User-defined function** → to calculate total engagement
  - **Stored procedure** → to fetch posts by username
- Improved **reusability and modularity**

✓ Automation using Trigger

- Implemented a **trigger** to automatically generate notifications when a post receives a like

7 Analysis and Queries

1. User Filtering

Query: Retrieve users whose names start with 'a'

```
select * from users

where user_name like 'a%';
```

Analysis: This helps identify specific users based on naming patterns and demonstrates the use of filtering conditions.

2. Posts Within Date Range

Query: Retrieve posts between specific dates

```
select * from posts

where posted_at between '2024-01-01' AND '2024-12-31';
```

Analysis:Used to analyze posts within a time period, useful for tracking yearly activity.

3. Top Active Users

Query: Users with highest number of posts

```
SELECT user_id,COUNT(post_id) as No_of_posts FROM posts

GROUP BY user_id

ORDER BY No_of_posts DESC

LIMIT 5;
```

Analysis: Identifies the most active users based on posting frequency.

4. Posts Per Day

Query: Count posts per day

```
select date(posted_at) as post_date, count(post_id) as total_posts from posts

Group by post_date  Having total_posts > 1;
```

Analysis: Helps understand daily activity trends and peak posting days.

5. Post Engagement (Likes & Comments)

```
select p.post_id,u.user_name,p.caption,count(distinct l.like_id)as 'total_likes',count(distinct
c.comment_id)as 'total_comments'

from posts p

JOIN users u ON u.user_id=p.user_id

LEFT JOIN comments c ON c.post_id=p.post_id

LEFT JOIN likes l ON l.post_id=p.post_id

GROUP BY p.post_id,u.user_name,p.caption;
```

Analysis: Provides a complete view of post performance based on likes and comments.

6. User Engagement Function

```
SELECT user_name, GetUserEngagement(user_id) AS total_engagement
FROM users
ORDER BY total_engagement DESC;
```

Analysis:Measures overall user engagement by combining likes and comments using a custom function.

7. Viral Posts (High Engagement)

```
SELECT * FROM post_summary;

SELECT post_id,user_name,caption,total_likes + total_comments Total_engagement

FROM post_summary

HAVING Total_engagement >

(SELECT AVG(total_engagement_val) FROM

(SELECT post_id,total_likes + total_comments total_engagement_val FROM post_summary HAVING
total_engagement_val <> 0) average_engagement);
```

Analysis: Identifies posts with higher-than-average engagement, indicating viral content.

8. Followers Analysis

Query: Users with highest followers

```
SELECT
    u.user_name,
    COUNT(f.follower_id) AS total_followers
FROM followers f
JOIN users u ON f.following_id = u.user_id
GROUP BY f.following_id, u.user_name
ORDER BY total_followers DESC;
```

Analysis: Identifies popular users based on follower count.

9. Trending Hashtags

```
SELECT
    h.tag_name,
    COUNT(ph.post_id) AS total_uses
FROM hashtags h
JOIN post_hashtags ph ON h.tag_id = ph.tag_id
JOIN posts p ON ph.post_id = p.post_id
WHERE p.posted_at >= DATE_SUB('2025-03-25', INTERVAL 30 DAY) -- Matching your 2025 data
GROUP BY h.tag_id, h.tag_name
ORDER BY total_uses DESC
LIMIT 5;
```

Analysis: Identifies the most frequently used hashtags in recent posts, helping detect trends.

8. Insights & Findings

- User engagement is driven more by content quality and relevance than posting frequency
- A few users emerge as top influencers, generating the highest likes and comments
- Not all active users receive high engagement, showing a gap between activity and impact

- · Posts with higher likes and comments are identified as viral or high-performing content
- · Hashtags like **#creativethinking** are frequently used, indicating trending topics
- · Follower distribution is uneven, with some users having higher influence and audience reach
- · User activity varies across days, helping identify peak engagement periods

· CONCLUSION

- · Successfully designed and implemented a relational social media database system using SQL
- · Established proper relationships using primary keys and foreign keys to maintain data integrity
- · Performed data analysis using joins, aggregations, subqueries, and functions
- · Identified key insights such as top influencers, active users, and trending hashtags
- · Evaluated post performance and user engagement effectively
- · Implemented stored procedures and user-defined functions for reusable logic
- · Used triggers to automate notifications, simulating real-time social media behavior
- · Transformed raw data into meaningful insights for decision-making
- · Demonstrated practical application of SQL in real-world social media analytics